Calculation of time and energy loss in transient operators

Starting, braking, speed change and speed reversal are transient operations. Time taken and energy dissipation in the motor during the transient operation can be evaluated by solving the motor load system torque equation, based on the nature of load torque i.e.

$T = \mathrm{T_L} + J\,\frac{d\omega_m}{dt}$ … (1) or $T = \mathrm{T_L} + J\,\frac{d\omega_m}{dt} + \mathrm{B}\omega_\mathrm{m}$ ….(2)

- When T and T_L are constants or proportional to speed, eq. (1) and eq. (2) will be a first-order linear differential equation, then it can be solved analytically. When T or T_L is neither constant nor proportional to speed, equation (1) and eq. (2) will be a non-linear differential equation, it could then be solved numerically by the Runge-Kutta method.
- Theoretically, transients are over in infinite time, which is not so in practice. In order to resolve this, transient operation is considered to be over when 95% change in speed has taken place.

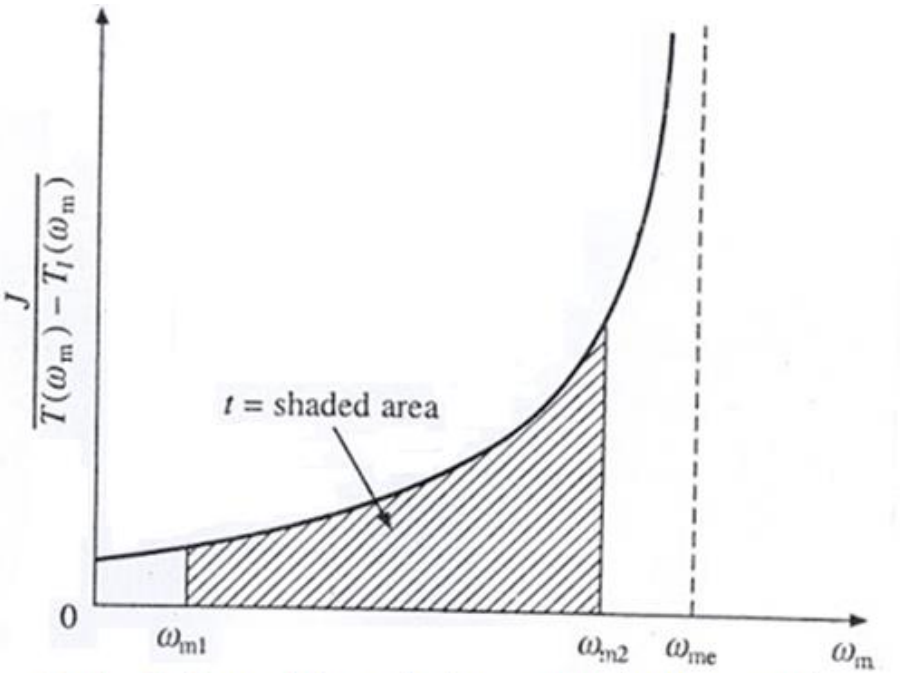
When speed changes from ω_{m1} to equilibrium speed ω_{me} , time taken for the speed to change from ω_{m1} to $[\omega_{m1} + 0{,}95\ (\omega_{me} - \omega_{\mathrm{m1}})]$ is considered to be equal to the transient time. the transient time can be calculated as,

$$dt = \frac{J\, d\omega_m}{T(\omega_m) - T_l(\omega_m)} \quad \ldots\ldots \quad (3)$$

The time taken for drive speed to change from ω_{m1} to ω_{m2} is obtained by integrated eq.(3)

$$t = \int_{\omega_{m1}}^{\omega_{m2}} \frac{J\, d\omega_m}{T(\omega_m) - T_l(\omega_m)} \quad \ldots\ldots \quad (4)$$

Eq.(4) can be integrated only if function $T\ (\omega_m)$ and $T_L(\omega_m)$ are known and are of integral form, otherwise the integral is evaluated graphically as shown below :



Caalculation of time during a transient operation

- The transient time can be evaluated by measuring the shaded area.
- When $\omega_{m2}$is in equilibrium speed ω_{me} ,then the reciprocal (مقلوب)of acceleration will become infinite at ω_{me} , time evaluated this way will be infinite. Transient time is computed by measuring the area between speed ω_{m1} and $\omega_{m1} + 0{,}95\ (\omega_{me} - \omega_{m1})$.
- Energy dissipated in a motor winding during a transient operation is

$$E = \int_0^t R.i^2\, dt$$

where ,
R: The motor winding resistance, i: Currant flowing through it.

Examples

Example 1: A drive has the following parameters:

J=10kg.m² , T=100-0.1N, N.m , Passive load torque TL=0.05N, N.m.

Initially, the drive is operating in a steady state. Now it is to be reversed. For this motor characteristic is changed to T= -100- 0.1N (N.m), calculate the time of reversal.

Solution:

For steady state:

$T=T_L$, or $100 - 0.1N_1 = 0.05N_1$, or

$0.15N_1=100$ then $N_1=\frac{100}{0.15}$= 666.7 rpm

- Theoretical speed after reversal for steady state speed, noting that the load is passive, $T=T_L$,or

$-100 - 0.1N_2 = 0.05N_2, N_{me} = N_2=\frac{-100}{0.15}$=-666.7 rpm

- Practical speed (considering that the motor starts reversing from zero rpm)

$N_2=N_1+0.95(N_{me}-N_1)$ or $N_2=0+0.95(-666.7-0)= -633.4$rpm

- During reversal T=-100-0.1N, TL=0.05

$$\mathcal{J}.\frac{d\omega}{dt} = T - T_L, \quad \text{or} \quad 10.\frac{d(\frac{2\pi N}{60})}{dt} = -100 - 0.1N - 0.05N.$$

So, $10 * \frac{2\pi}{60} * \frac{dN}{dt} = -100 - 0.15N$ or

$\frac{dN}{dt} = \frac{3}{\pi}(-100 - 0.15N) = -95.49-0.143$ N or

$$dt = \frac{dN}{-95.49 - 0.143N}$$

- Integrating on both sides we get

$$\int_0^{t_r} dt = \int_{666.7}^{-633.4} \frac{dN}{-95.46 - 0.143N}$$

So, The time required for the reversal of drive is

$$t_r = \frac{1}{-0.143}\left[\ln(-95.49 - 0.143N)\right]_{666.7}^{-633.4}$$

$$= \frac{-1}{0.143}\left[\ln\left(-95.49 - 0.143 * (-633.4)\right) - \ln(-95.49 - 0.143(666.7))\right]$$

$$= \frac{-1}{0.143}\left[\ln(-4.9138) - \ln(-190.8281)\right]$$

$$= \frac{-1}{0.143}\left[\ln\left(\frac{-4.9138}{-190.8281}\right)\right] = \frac{-1}{0.143}(-3.66) = 25.58 \text{ sec}$$

- Example 2:

Calculate the starting time of a drive with

the following parameters:

$J = 10\ Kg_{m^2}, T = 15 + 0.5\ \omega_m \quad and\ T_L = 5 + 0.6\ \omega_m$

Solution:

- At a steady state, fundamental torque equation is

$T = T_L$. So,

$15 + 0.5\ \omega_{me} = 5 + 0.6\ \omega_{me}$, or $0.1\omega_{me} = 10$

and finally , $\omega_{me} = 100\ rad/sec$

- Consider final speed $\omega_2 = 0 + 0.95(100 - 0)$= 95 rad/sec

To starting, time of drive is time taken for drive speed from 0 rad/sec to 95rad/sec

- The fundamental torque equation at transient of starting is

$\mathrm{T} = \mathrm{T}_L + J\frac{d\omega}{dt}$ or $15 + 0.5\omega = 5 + 0.6\omega + 10\frac{d\omega}{dt}$, then

$$dt = \frac{10}{10 - 0.1\omega}\, d\omega$$

- Integrating on both sides for starting time is

$$\int_0^{\mathrm{ts}} dt = \int_0^{95} \frac{10}{10 - 0.1\omega}.d\omega$$

- $ts = 10\left[\frac{\ln(10-0.1\omega)}{-0.1}\right]_0^{95} =$

$$-100[\ln(10 - 0.1 * 0.95) - \ln(10 - 0)]$$

$$= -100[\ln(0.5) - \ln(10)] = -100\ln\left(\frac{0.5}{10}\right)$$

$$= -100\ln(0.05) = 2.9975 \text{ sec}$$

- The starting time for the drive is, ts = 2.9975 sec